AgenticRotation: Supplementary Material

Anonymous CVPR submission

A. Scope of This Release

This supplement accompanies the working manuscript dated 29 October 2026. Its empirical results concern the marked RSLCube and a single frozen pose-latent policy release. The archive contains 400 unique trials: 80 physical diagnostics and 320 semantic trials. The full Astra condition appears in several comparisons but is counted once. Results from an earlier RGB-D backend, invalid contact-capacity runs, and development seeds are not pooled with this release. Missing experiments are marked TODO throughout. No synthetic performance numbers are used.

The original eight-condition suite has 320 trials. The two later bare VLM conditions add 80. The latter were requested after the original evaluation to distinguish complete-system baselines from shared-controller baselines. Their addition changes neither the physical metric nor the selected weights. Since the condition definitions were amended after earlier outcomes existed, the full study is not described as a blindly preregistered comparison.

B. Exact Task and Runtime Contract

B.1. Object and goals

Table 1 records the fixed face mapping. Body normals include the original mesh rotation. All methods use the same mapping and cached six-vector bank. The cyan channel corresponds visually to a pale-blue face and magenta to a purple face. The target quaternion gives a canonical twist as well as the upward face normal.

Letter	Skill color	Body normal	Target XYZW
T	red	$(0, -1, 0)$	$(-s, 0, 0, s)$
E	green	$(-1, 0, 0)$	$(0, s, 0, s)$
R	blue	$(0, 0, -1)$	$(1, 0, 0, 0)$
L	yellow	$(0, 0, 1)$	$(0, 0, 0, 1)$
S	magenta	$(0, 1, 0)$	$(s, 0, 0, s)$
H	cyan	$(1, 0, 0)$	$(0, -s, 0, s)$

Table 1. Frozen goal mapping, with $s = \sqrt{1/2}$.

The semantic instruction identifies a letter. It does not specify an arbitrary target quaternion. The benchmark scores its associated canonical quaternion at a 0.16-radian tolerance. A drop is recorded when the distance from the object's nominal grasp position is at least 0.24 metres. This is an operational benchmark termination, not a detector of every transient slip or lost contact. The object may move within that boundary without a recorded drop.

B.2. Physics and perception

The runtime uses Isaac Lab 3.0, Isaac Sim 6.2, Newton 1.6.1, Warp 1.17, and PyTorch 2.11 with CUDA 13.0. The validated collision model contains 23 bodies and 117 shapes, comprising 116 convex shapes and one box. The contact allocation uses `nconmax=4096` and `njmax=1024`. Earlier overflow runs are invalid even if their reported rewards appear plausible.

The pose estimator is frozen. It predicts 27 image-coordinate/depth outputs for nine points, then recovers pose through calibrated geometry. The object image is rendered at 120×120 with live hand occlusion. Sensor calibration, object asset, policy normalization, joint order, action mapping, and smoothing belong to the release contract. A change to one of these produces a new release. The current setup does not establish real-camera robustness or unseen-scene generalization.

B.3. Timing, termination, and exclusions

The semantic horizon is 120 seconds of logical control time. Model latency is measured independently and translated into a minimum pending duration of $\lceil 30L \rceil$ steps. The request must also have actually completed. Physics is never intentionally paused to await a semantic response. Because simulation throughput is below the logical 30 Hz rate, wall time per episode exceeds logical duration. A hardware claim requires a separately measured end-to-end rate.

The shared gate runs at 3 Hz. It requires an accepted requested-symbol binding, a non-inspection command, no outstanding request, and estimated canonical orientation error no greater than 0.16 radians. The independent scorer applies the same angular tolerance to true orientation and checks the actual requested face. No post-stop dwell is required. Success is not the minimum orientation error at any earlier time.

A service error or model call exceeding 120 seconds is recorded without replacement. The main three-system panel removes trial 183102011 from every arm. The four-

way control panel removes 183102000, 183102010, and 183102011. The resulting denominators are 39 and 37. Their percentages are not directly interchangeable. Raw 40-trial outcomes remain visible, including service-affected cases.

C. Models and Training

C.1. Semantic and motor inputs

The motor policy receives 109 target-free proprioceptive/history features, seven measured pose values, and a 768-dimensional goal. Its goal projection has width 64 and a tanh nonlinearity. The inherited recurrent policy emits 21 normalized joint targets. The asset-specific goal quaternion is absent from the migrated student's inference inputs.

The hosted model route recorded in the manifests is `gpt-6-astra`. The local grounding backbone is Qwen3-VL-4B-Instruct. The Astra pathway uses public tool actions and a deterministic goal-bank lookup. The fine-tuned pathway predicts a continuous vector within the normalized skill span. Neither pathway receives the requested face's true pose. The local unreadable-view schedule is $(1, 3, 0, 2, 4, 5)$ with 90-step inspections. Astra chooses inspection actions of 10–90 steps. This scheduler difference is part of the complete-system comparison and a confound for attributing all effects to model weights.

[TODO: Document the hosted model's exact serving revision, sampling settings, and full sanitized system prompt/tool schema in the final anonymous release. A route name alone does not guarantee reproducibility across provider updates.]

C.2. Grounding supervision

The grounding model first uses 4,096 teacher-view training examples and 1,024 validation examples, then 4,800 student-view training examples and 2,400 validation examples with mixed replay. Each adaptation stage uses 240 updates. Training and validation collection seeds are disjoint from the formal evaluation seeds. Readability is a geometric/image-signal proxy. The objective combines readability prediction, readable-example skill classification, and cosine alignment to a canonical vector. We do not claim human-verified text recognition labels or an action-consistency objective in this release.

The fixed layout permits learned letter-to-color shortcuts. A held-out pose or trajectory alone does not remove that possibility. A convincing grounding test changes or hides the visual evidence while holding the instruction and physical state fixed, then checks whether the model's selected target changes appropriately.

C.3. Controller features and losses

The sensor branch uses the current proprioceptive vector, measured pose, student belief and proposal, request-anchor pose and action, fitting residual, binding flag, pending flag, and elapsed pending duration. Its normalized inputs are clipped to $[-8, 8]$. Goal features are centered and scaled using the fixed bank. The sensor, goal, and fusion widths are 192, 64, and 128. Each actor has 22 outputs: a speed logit and 21 residual values.

The waiting actor interpolates from current joint positions toward the proposal and adds a bounded residual. The execution actor applies a bounded residual to the proposal at full speed. Both are supervised from the teacher with residual regularization of 0.01. Waiting labels hold the request anchor. Execution labels follow the active goal. Paired state/goal examples and schedule-derived states are used during fitting. Their causal utility remains to be established by matched training ablations.

No raw RGB, true object pose, teacher action, future model response, or oracle semantic label is a deployed actor input. Vision affects the actor through estimated pose and recurrent belief. The reported stopping gate is measured-pose based. Previously trained completion-evidence heads are inactive and cannot be credited for the current results.

D. Additional Numerical Results

Semantic condition	Seed 1	Seed 2	Raw total
Astra full	13/20	16/20	29/40
Fine-tuned VLM, bare	14/20	13/20	27/40
Rule VLM, bare	5/20	9/20	14/40
Fine-tuned VLM, shared control	15/20	15/20	30/40
Rule VLM, shared control	6/20	10/20	16/40
Astra, fixed slow / student	11/20	12/20	23/40
Astra, learned wait / student	15/20	11/20	26/40
Astra, fixed slow / learned exec.	16/20	12/20	28/40

Table 2. Raw semantic successes by seed. Seed 1 is 183101 and seed 2 is 183102. The bare addendum contributes the second and third rows only.

Comparison on eligible trials	A-only	B-only	Exact p
Full Astra / bare fine-tuned	9	6	0.6072
Full Astra / bare rule	18	2	0.0004
Full Astra / fixed control	15	8	0.2100
Full Astra / waiting only	8	6	0.7905
Full Astra / execution only	8	7	1.0000

Table 3. Exploratory two-sided exact McNemar tests. The first two rows use 39 trials and the last three use 37. Tests are unadjusted and do not establish all module effects.

On raw paired trials, adding shared control to the fine-tuned VLM changes successes from 27 to 30 and drops from nine to five. Five trials succeed only without the actors and eight only with them, giving $p = 0.581$. The rule VLM changes from 14 to 16 successes, with $p = 0.754$. These auxiliary results must remain available even though the primary baselines omit the actors. They show that controller access can affect a complete-system comparison and that current evidence does not establish an Astra-specific benefit.

Individual rate intervals use the Wilson method with nominal 95% coverage. Paired comparisons use trial identities and initial-state verification. They account for pairing but do not address all uncertainty from hosted model variability or future object distributions. A single two-seed cohort is insufficient to establish a stable ordering between systems separated by two or three raw successes.

E. Successor Student: A Separate Development Study

The existing cube results use a fixed student checksum. Successor training is designed to ask whether extra DAgger with the actual frozen CNN sensor improves physical control beyond an equal budget with simulated pose noise. Both branches start from that same latent student, use training seed 183201, and run 400 updates with 16 environments, 45 rollout steps, and four fitting epochs. Learning rate is 3×10^{-4} with a new Adam state and zero weight decay. The teacher action fraction decays as $0.9 \cdot 0.95^{i}$. Each branch collects 288,000 transitions. The CNN and teacher remain frozen.

After training, the final checkpoint from each branch is compared with the released student and the frozen latent student. Every arm uses CNN observations and 40 fresh diagnostic trials from seeds 183221 and 183222. No intermediate checkpoint is selected from those trials. The CNN branch must exceed both latent references in reach count without increasing drops to pass the development admission rule. This gate is an engineering selection rule, not a significance claim.

A selected new student still requires separately collected controller data and compatibility validation. Only then may a new semantic release use reserved seeds 183301 and 183302. The original cube-v1 results remain unchanged. Both branches and all 160 diagnostic trials have completed with valid physics logs. The four reach counts are 39/40, 38/40, 39/40, and 40/40 for the released, frozen latent, extra simulated, and extra CNN policies. Drops are one, two, one, and zero. The CNN candidate passes the stated gate and has not been automatically deployed. Exact paired tests against the frozen latent and extra simulated policies give $p = 0.5$ and $p = 1.0$. Thus this is a promising development result, not an established perception-matching effect.

F. Separate Follow-up Cohorts

These studies retain the frozen task success criterion and do not replace the completed cube-v1 cohort. Service-eligible success is the primary metric. Cross-method comparisons remove the same service-affected slots from every arm. All raw records, including excluded physical drops, remain available. Forty scheduled slots do not necessarily yield forty eligible trials.

F.1. Completed independent replication

Seeds 183411 and 183412 each provide 20 trials per method. All 120 scheduled records pass the source, checkpoint, physical-log and initial-state audits. Slots 183411008, 183411015 and 183411017 have recorded Astra service errors and are removed from every arm for the matched comparison. The last includes a connection failure after approximately 126 seconds and a waiting-phase drop. Its exclusion follows the existing service rule. Raw success counts are 26/40, 29/40 and 22/40 for Astra, bare fine-tuned VLM and bare rule VLM. The common eligible counts appear below.

Method	Eligible success	Rate (%)
Astra full	26/37	70.3
Fine-tuned VLM, bare	27/37	73.0
Rule VLM, bare	21/37	56.8

Table 4. Completed independent replication on 37 common service-eligible initial states from 40 scheduled trials per method. This supersedes the archived partial snapshot and remains separate from the original cube-v1 cohort.

The eleven eligible Astra failures comprise nine drops, one false stop and one abstention. The drops include one during startup, two during subsequent waiting and six outside waiting. Raw Astra counts additionally include one service-excluded drop. Against the fine-tuned VLM, Astra has five exclusive successes and six exclusive failures (exact paired $p = 1.0$). Against the rule VLM, the counts are eleven and six ($p = 0.332$). These exploratory comparisons do not establish superiority. The earlier 18-slot snapshot remains archived for provenance but is no longer used for the follow-up table.

F.2. Control interaction and explicit goals

The completed factorial retains the common 37-trial panel shown in the main paper. Learned waiting alone reduces startup drops from four to one but introduces three later waiting drops, leaving total waiting-related drops at four. With execution recovery present, learned waiting eliminates four startup drops while execution drops increase from four to nine. The full method wins eight trials lost by execution-only control and loses seven won by that control. Hence

net success improves by one, without a resolved interaction benefit.

A deterministic restoration diagnostic verifies 19 of 20 planned physical states. Each state has two assigned goals and both actor conditions, yielding 76 valid branches. State 12 cannot be restored because its prefix terminates, and is not replaced. Both conditions give the assigned goal to the student. No waiting drops occur, but their subsequent execution differs.

Actor input	Final in tolerance	Drops
Explicit actor goal	26/38	5
No explicit actor goal	29/38	2

Table 5. Paired recovery diagnostic on 19 states, two goals each. Four branches in each condition already start within tolerance. Final tolerance is not autonomous semantic success. Explicit actor-goal conditioning is not supported as beneficial by this diagnostic.

Mean assigned-goal progress is 1.195 radians with explicit actor-goal input and 1.340 without it. Different first actions under different goals do not isolate the actor's explicit goal pathway because the student itself is goal-conditioned. The separately trained correct-goal, no-explicit-goal and shuffled-goal semantic comparison remains incomplete.

F.3. Completed bare-VLM visual interventions

The upper-level image changes while the low-level pose predictor continues to receive the original image. Lighting is fixed. The permuted condition assigns the next symbol in (T, E, R, L, S, H) to each color face and changes the requested symbol accordingly, preserving the scored physical target across conditions. The masked condition hides the lettering. Geometry-consistent image construction and the transmitted image hashes are audited. Seeds 183441 and 183442 provide 20 trials each, and physical initial-state hashes match across the three conditions.

Method	Identity	Permuted	Masked
Fine-tuned VLM	30/40	16/40	3/40
Rule VLM	23/40	23/40	0/40
Astra full	TODO	TODO	TODO

Table 6. Completed upper-image interventions for the bare VLMs. Every numerical cell contains 40 service-eligible trials, with no service exclusions. Astra cells are pending, not zero. These cohorts are separate from the main cube-v1 table.

Fine-tuned success falls by 35 percentage points after remapping and to 7.5% after masking. Rule-based success is 57.5% under both identity and remapping, then zero after masking. Equal aggregate identity and remapped rates do not imply identical trial outcomes. The results establish intervention sensitivity but do not show that all fine-tuned successes arise from a fixed lookup, or that Astra is more robust before its matched arms finish.

Applied semantic bindings. We audit every applied, accepted call from the same 240 trials. The episode-file hashes match the archived results, and the last accepted skill agrees with the recorded terminal binding. Table 7 separates acquiring the correct target from completing it. Under permutation, the fine-tuned model first binds correctly in 25 trials and incorrectly in eleven, compared with 34 and four under identity. Nine of its eleven wrong-binding trials select the original color for the newly requested symbol. The rule model has six wrong-binding trials under each layout, with three following the old association after permutation. This pattern is consistent with reliance on training associations, but does not identify the model's internal mechanism.

Model	Layout	Correct	Wrong	None	Old
Fine-tuned	Identity	34	4	2	–
Fine-tuned	Permuted	25	11	4	9
Fine-tuned	Masked	3	1	36	–
Rule	Identity	24	6	10	–
Rule	Permuted	25	6	9	3
Rule	Masked	0	0	40	–

Table 7. First accepted binding in each of 40 trials. Old counts trials with an accepted skill matching the original symbol–color association, defined only for permutation. These are the same trials as the success table, not new evaluations.

No wrong-to-correct binding transition occurs in these bare-VLM cohorts. This is descriptive: trajectories, visibility, and opportunities for additional calls differ. Of fourteen fine-tuned false stops under permutation, ten end with a wrong binding and four with the correct binding but an independently incorrect final pose. Masking leads to no accepted binding in 36 fine-tuned trials and all forty rule-model trials. The three masked fine-tuned successes cannot establish letter grounding when that information is removed.

Identity versus permutation yields eighteen exclusive successes and four exclusive failures for the fine-tuned model (exact paired $p = 0.00434$), versus ten and ten for the rule model ($p = 1.0$). These are exploratory paired comparisons without multiplicity correction. The single deterministic remapping and edited upper images limit generalization claims. Astra's corresponding calls and recovery behavior must be audited after its cohorts complete.

F.4. Outstanding agent feedback evidence

The same-Astra comparison uses one initial bounded plan versus at most six feedback calls. The student, ac-

tors, stopping gate and physical horizon are held fixed. Protocol checks, synthetic downstream rollouts and real development-image interface checks have passed. Formal results for the two 40-trial arms are pending. No benefit from subsequent observation or replanning is inferred from interface validation alone.

G. Experimental Requirements Before Submission

G.1. Separate model strength from control

The primary comparison uses full Astra against bare VLM baselines. Its interpretation is system-level. To attribute gains to the upper-level model, compare all providers with identical control actors, timing conventions, stopping, and information access. Preserve the existing shared-control results. Repeated live calls can produce different public plans even on matching initial states. Fixed-public-plan replay can isolate motor changes, but replay success must not be reported as a new live Astra success rate.

G.2. Test semantic and visual causality

Use the same physical state with different valid requested faces. Measure action direction and completion with the correct goal, a shuffled goal, and no goal. Train the no-goal actor with equal capacity and data. Permute the letter-color layout or mask/shuffle image evidence while keeping background and lighting fixed. Pair these with a text-only baseline. These controls determine whether success depends on current visual semantics rather than a fixed lookup.

G.3. Test agent feedback

Hold the model, student, and local actors fixed. Compare one initial semantic query with the complete feedback loop, then remove history or verification feedback. Keep the registered one-call and at-most-six-call budgets explicit and record actual calls, logical duration, wall time, and service failures. This contrasts complete protocols and does not isolate fresh information from extra inference compute. Latency replay should use the same public plan and delay sequence across motor variants. A simple time-aware residual actor provides a stronger learning baseline than fixed slowing alone. It should not be labeled an official reproduction of A2C2 or ARLI without a verified implementation.

G.4. Separate perception accuracy and control benefit

Compare the original and improved predictors with the same student and agent first. Report independent translation/rotation error, tail error, keypoint visibility, false stops, and semantic success. Oracle pose estimates the residual motor gap. Only a second experiment should adapt the student to the new predictor. A lower pose error is useful evidence, but it does not guarantee a higher task success rate.

G.5. Objects and physical hand

Select about five objects with variation in geometry, inertia, contact affordances, and semantic ambiguity. Record object-specific students and training budgets. Balance the task set within each object and report each row before computing an object macro-average. Forty trials per arm per object is the default batch size. Hardware runs need calibrated independent scoring, documented resets, two sessions, and synchronized video and command logs. A new object checkpoint alone is insufficient without its assets, calibration, normalization, action conventions, and goal definitions.

H. Reproducibility and Evidence Audit

The accompanying research archive stores 400 frozen-release trial records and 160 separately labeled successor development records, sanitized summaries, source-file checksums, and checkpoint hashes. Runtime hostnames and authorization material are excluded. Original local evidence is retained separately and hashed before export. Result tables and plots are regenerated from the archived data. A separate follow-up archive contains 120 completed independent-replication records, 240 completed visual-intervention trials, and summaries of the 19-state recovery diagnostic. The historical 98-record partial snapshot is retained separately and adds no trials. The replication is not pooled with the original cohort to form a revised main result. The public goal mapping and physical success contract are frozen for cube-v1. New predictors, students, objects, or stop rules require separate versioned releases.

[TODO: Complete the anonymous implementation package and object-specific licenses before submission.] The manuscript itself contains no link to an identifying project repository. Public repository publication is a development convenience, not a replacement for a self-contained anonymous submission.